Number 8 — Converting units of length, area and volume · Compound measures

Calculator allowed. Marks in brackets. ★ = harder question. Give answers to 3 s.f. where rounding is needed.

1 A photograph measures 15 cm by 10 cm. Find its area in m². [2]

2 One litre is 1000 cm³. How many cubic millimetres are there in 2.5 litres? Give your answer in standard form. [3]

3 Sound travels at 340 m/s in air. **a** How many kilometres does sound travel in one day? **b** A cube has sides of that length. Work out its volume in km³, in standard form to 3 s.f. [3]

4 A person drinks about 730 litres of water a year. What is this in millilitres per minute? [2]

5 **a** A cyclist rides for 25 minutes at an average speed of 8 m/s. How far does she ride, in kilometres? **b** A sprinter runs 60 m in 7.5 s. What is his speed in km/h? [6]

6 **a** A brass ornament has density 8.5 g/cm³ and mass 1.7 kg. Find its volume. **b** A cubic metre of oak has a mass of 720 kg. Find the density of oak in g/cm³. [6]

7 Pressure $= \dfrac{\text{force}}{\text{area}}$. Copy and complete the table. [3]

force	area	pressure
60 N	2.5 m²	___ N/m²
___ N	4.8 m²	12.5 N/m²
1500 N	___ m²	300 N/m²

Total: 25 marks

Check: area units scale by the square of the length factor (1 m² $= 10^4$ cm²), volume units by the cube (1 m³ $= 10^6$ cm³ $= 10^9$ mm³) · m/s → km/h: multiply by 3.6 · match units before dividing (g with cm³, kg with m³) · density = mass ÷ volume, pressure = force ÷ area — rearrange with a formula triangle.